Supplementary material

Decision support for durian (*Durio zibethinus*) disease diagnosis from field photographs: what grouped validation shows about backbone choice and referral

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All tables and figures are generated by `paper/make_supplement.py` from the released result files (`results/preds.csv`, `results/results.csv`, `results/v2/results.csv`, `results/v3/*`, `results/numbers.json`, `paper/numbers.json` and `paper/numbers_v3.json`). The single-channel spatial gate (SG) is labelled `lfa` / LFA in the code, the result files and Figs. S1–S2, and the leaf blight complex is labelled `Leaf_rot`.

Table S1. Per-class precision, recall and F1 (%, mean ± SD over runs in which the class was present), all test images

Class	Configuration	Precision	Recall	F1
Algal leaf spot	None	72.9 ± 25.5	63.7 ± 18.4	65.3 ± 19.3
	SG	73.8 ± 25.9	61.1 ± 21.2	63.6 ± 21.0
	SE	72.8 ± 25.2	59.9 ± 20.6	62.5 ± 19.9
	CBAM	74.1 ± 24.1	60.7 ± 21.8	63.6 ± 21.1
Leaf blight complex	None	66.6 ± 15.4	78.9 ± 16.0	70.7 ± 12.3
	SG	66.9 ± 16.2	82.9 ± 15.1	71.8 ± 11.1
	SE	64.8 ± 17.9	79.3 ± 20.1	69.4 ± 15.5
	CBAM	66.0 ± 14.5	82.2 ± 12.3	71.8 ± 9.9
Phomopsis leaf spot	None	77.2 ± 20.5	78.6 ± 19.0	74.2 ± 13.4
	SG	77.7 ± 20.2	76.7 ± 19.4	73.5 ± 13.9
	SE	78.3 ± 20.9	78.9 ± 18.8	74.9 ± 14.7
	CBAM	77.0 ± 21.0	78.1 ± 20.4	73.4 ± 14.9
Root and collar rot	None	91.0 ± 18.3	99.3 ± 2.1	93.6 ± 14.2
	SG	92.3 ± 16.2	99.0 ± 2.2	94.6 ± 11.6
	SE	91.2 ± 18.3	99.3 ± 1.5	93.8 ± 14.2
	CBAM	91.5 ± 17.3	99.2 ± 2.0	94.0 ± 13.1

Table S2. Accuracy (%, mean ± SD over 40 runs) and paired difference from the plain network over 20 folds (corrected resampled *t*-test)

Configuration	Accuracy	Difference (pp) [95% CI]	*p*, corrected	Folds higher
None	75.95 ± 9.33	–	–	–
SG	75.47 ± 10.09	−0.49 [−4.67, 3.70]	0.811	9/20

Configuration	Accuracy	Difference (pp) [95% CI]	*p*, corrected	Folds higher
SE	74.75 ± 10.93	−1.20 [−8.79, 6.39]	0.744	10/20
CBAM	75.78 ± 9.66	−0.17 [−4.82, 4.47]	0.939	10/20

Table S3. Preliminary experiment: eight runs per configuration on one fixed five-class split (earlier code; training randomness not controlled by the seed, so replicates are unpaired)

Configuration	Macro F1, mean ± SD	Difference from none (pp)	*p*, Welch *t*	*p*, Mann–Whitney
None	77.55 ± 1.91	–	–	–
SG	78.05 ± 2.61	+0.50	0.671	0.958
SE	76.63 ± 3.75	−0.92	0.551	0.752
CBAM	77.10 ± 2.86	−0.45	0.718	0.833

Table S4. Macro F1 averaged over all four classes (an absent class scored as zero), as stored by the training notebooks and used in the pre-specified decision rule

Configuration	Macro F1 (%), mean ± SD	Paired difference (pp) [95% CI]	*p*, corrected	Meets rule (≥ 2 pp and CI > 0)
None	72.44 ± 13.05	–	–	–
SG	72.32 ± 13.08	−0.12 [−4.54, 4.30]	0.954	no
SE	71.63 ± 13.45	−0.82 [−7.63, 6.00]	0.805	no
CBAM	72.18 ± 13.43	−0.27 [−5.11, 4.57]	0.909	no

Table S5. Training settings (identical for all configurations)

Item	Setting
Backbone	EfficientNet-B0, torchvision ImageNet weights (IMAGENET1K_V1)
Input	Train: RandomResizedCrop(224, scale 0.7–1.0). Evaluation: Resize(256) on the short side, CenterCrop(224)
Augmentation	Horizontal flip p = 0.5; vertical flip p = 0.3; RandomAffine (rotation ±20°, translation 5 %, scale 0.95–1.05); ColorJitter (brightness 0.3, contrast 0.3, saturation 0.2, hue 0.05); GaussianBlur (kernel 3–7, σ 0.1–1.5); ImageNet normalisation; RandomErasing p = 0.3, 2–10 % of the image
Classifier head	Dropout p = 0.3, Linear(1280, 4)
Stage 1	15 epochs, backbone frozen (BatchNorm statistics still updating), Adam, lr 1e-3, attention module + head trained
Stage 2	15 epochs, all parameters trainable, Adam, lr 1e-5, starting from the best stage-1 checkpoint
Checkpoint selection	Highest inner-validation macro F1 over both stages; inner validation ≈ 10 % of the training part, by whole groups
Loss and sampling	Cross-entropy with inverse-frequency class weights; WeightedRandomSampler with class weights
Batch size	16
Precision	Mixed precision (torch.autocast with a gradient scaler)
Seeds	0 and 1 per fold; Python, NumPy, PyTorch, sampler and data-loader workers seeded; cuDNN deterministic
Hardware	Google Colab GPU runtime; PyTorch and torchvision

Table S6. Paired comparisons of macro F1 with the plain network under three tests over the 20 folds

Configuration	Mean difference (pp)	*p*, corrected resampled *t*	*p*, uncorrected paired *t*	*p*, Wilcoxon signed-rank
SG	−0.13	0.953	0.870	0.898
SE	−0.77	0.816	0.521	0.956
CBAM	−0.38	0.879	0.675	0.277

Table S7. Sensitivity to the grouping of video frames (Section 2.3 of the manuscript)

Test images scored	Mean test images per fold	Configuration	Macro F1 (%), mean ± SD	Paired difference (pp) [95% CI]	*p*, corrected
All test images	137.5	None	75.06 ± 10.86	–	–
		SG	74.93 ± 10.95	−0.13 [−4.56, 4.31]	0.953
		SE	74.29 ± 11.80	−0.77 [−7.62, 6.08]	0.816
		CBAM	74.68 ± 10.78	−0.38 [−5.48, 4.73]	0.879
Video frames excluded	121.8	None	66.60 ± 12.04	–	–
		SG	66.46 ± 12.15	−0.13 [−4.55, 4.28]	0.950
		SE	65.34 ± 12.74	−1.26 [−10.28, 7.76]	0.773
		CBAM	66.37 ± 13.06	−0.23 [−5.83, 5.37]	0.933
Only images whose video-linked group was not split	87.0	None	67.36 ± 12.17	–	–
		SG	66.54 ± 12.34	−0.83 [−6.09, 4.44]	0.746
		SE	64.54 ± 14.16	−2.83 [−11.70, 6.04]	0.512
		CBAM	66.19 ± 14.53	−1.17 [−7.96, 5.62]	0.722

Without the video frames the mean seed difference was 5.67 points, and the SG was at least 3 points ahead of the plain network in 20.0 % of seed-matched pairs and at least 3 points behind in 22.5 %.

Table S8. Model size, computation and CPU latency for one 224 × 224 image (median and 90th percentile of 200 passes, Colab CPU)

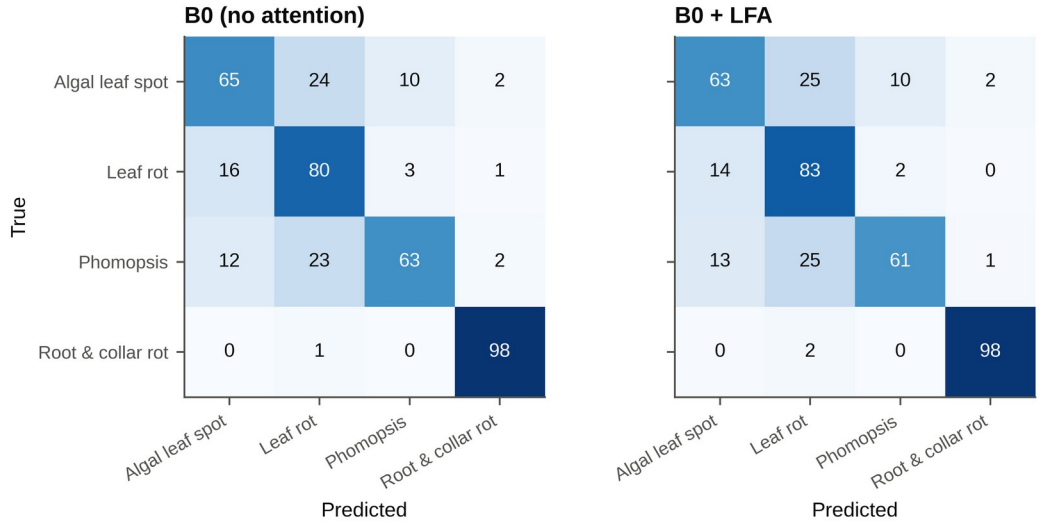
Configuration	Parameters (M)	Added parameters	GMACs	1 thread, median (ms)	1 thread, p90 (ms)	12 threads, median (ms)	12 threads, p90 (ms)
None	4.0127	0	0.3845	34.29	35.38	22.46	46.40
SG	4.0140	1,281	0.3846	34.18	35.39	22.65	23.92
SE	4.2175	204,800	0.3847	34.21	35.17	22.54	24.78
CBAM	4.2176	204,898	0.3850	35.20	35.86	23.29	27.03

Table S9. Fold sizes and macro F1 per fold (%, mean of two seeds; † no root and collar rot image in the test fold)

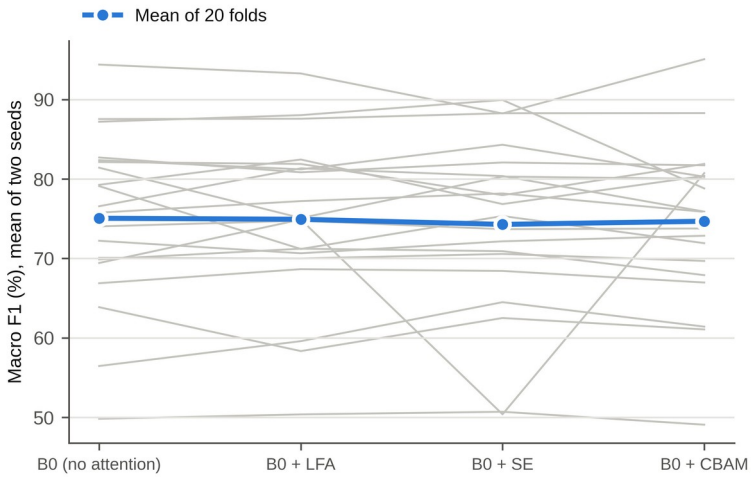
Repeat	Fold	Train	Validation	Test	None	SG	SE	CBAM
0	0	440	58	52	94.40	93.28	88.25	95.07
0	1	386	41	123	75.76	77.22	78.18	75.87
0	2	275	28	247	63.87	58.34	62.51	61.07
0	3	388	34	128	82.14	81.89	77.97	81.90

Repeat	Fold	Train	Validation	Test	None	SG	SE	CBAM
1	0	453	42	55	79.10	71.19	75.34	71.92
1	1	395	46	109	76.57	81.34	80.37	75.90
1	2†	296	30	224	49.82	50.38	50.71	49.08
1	3	348	40	162	74.04	74.75	73.69	73.77
2	0†	419	51	80	72.23	70.65	72.16	72.86
2	1	256	35	259	66.89	68.65	68.43	66.97
2	2	443	46	61	82.37	81.21	84.31	80.29
2	3	358	42	150	81.42	75.09	80.30	80.01
3	0	393	61	96	69.43	74.90	50.39	80.74
3	1	424	46	80	70.07	70.00	70.57	69.69
3	2	247	23	280	69.88	71.23	70.92	67.90
3	3	417	39	94	87.54	87.58	88.28	88.29
4	0†	459	50	41	87.21	88.04	89.95	78.82
4	1	381	34	135	82.71	80.84	82.08	81.72
4	2	309	31	210	56.47	59.59	64.50	61.42
4	3	342	44	164	79.27	82.45	76.84	80.39

Figures S1 and S2



Confusion matrices of the plain network and the SG (labelled LFA), pooled over the 40 runs of each configuration and normalised by row (%), all test images.



Macro F1 of each configuration on each of the 20 folds (grey lines, mean of two seeds) and the mean over folds (blue), all test images.

Table S10. Per-class precision, recall and F1 (%, mean ± SD over runs in which the class was present), video frames excluded (primary scoring)

Class	Configuration	Precision	Recall	F1	Runs
Algal leaf spot	None	73.1 ± 25.5	63.7 ± 18.4	65.4 ± 19.3	40
	SG	74.1 ± 26.0	61.1 ± 21.2	63.7 ± 21.1	40
	SE	73.1 ± 25.3	59.9 ± 20.6	62.5 ± 19.9	40
	CBAM	74.6 ± 24.4	60.7 ± 21.8	63.7 ± 21.1	40
Leaf blight complex	None	67.5 ± 15.8	78.9 ± 16.0	71.1 ± 12.3	40
	SG	68.0 ± 16.4	82.9 ± 15.1	72.5 ± 11.0	40
	SE	65.9 ± 18.5	79.3 ± 20.1	69.9 ± 15.5	40
	CBAM	66.7 ± 14.7	82.2 ± 12.3	72.2 ± 9.9	40
Phomopsis leaf spot	None	44.3 ± 44.2	29.4 ± 27.8	32.0 ± 31.6	32
	SG	44.4 ± 45.0	27.0 ± 27.3	30.9 ± 31.7	32
	SE	44.1 ± 44.0	31.0 ± 28.2	32.8 ± 32.3	32
	CBAM	44.2 ± 44.4	28.6 ± 28.2	31.4 ± 31.7	32
Root and collar rot	None	91.0 ± 18.3	99.3 ± 2.1	93.6 ± 14.2	34
	SG	92.3 ± 16.2	99.0 ± 2.2	94.6 ± 11.6	34
	SE	91.2 ± 18.3	99.3 ± 1.5	93.8 ± 14.2	34
	CBAM	91.5 ± 17.3	99.2 ± 2.0	94.0 ± 13.1	34

Table S11. Backbone comparison: macro F1 and accuracy (%, mean ± SD over 40 runs) and paired difference from EfficientNet-B0 over 20 folds

Corrected resampled *t*-test (factor 0.383); Holm adjustment over the four comparisons; "rule" is the pre-specified criterion (difference of at least 2 pp and Holm $p < 0.05$).

Scoring	Backbone	Macro F1	Accuracy	Difference (pp) [95% CI]	*p*, corrected	*p*, Holm	*p*, uncorrected *t*	*p*, Wilcoxon	Folds higher	Rule met	Mean seed difference (pp)
video frames excluded (primary)	EfficientNet-B0	66.65 ± 12.20	72.82	–	–	–	–	–	–	–	5.61
	MobileNetV3-Large	64.39 ± 14.42	72.15	−2.26 [−9.47, 4.95]	0.520	0.520	0.0851	0.1327	8/20	no	5.94
	MobileNetV4-Conv-Medium	58.30 ± 12.67	66.53	−8.36 [−16.38, −0.33]	0.042	0.126	< 0.0001	< 0.0001	2/20	no	7.42
	ConvNeXt V2-Tiny	74.81 ± 12.10	82.89	+8.16 [1.87, 14.45]	0.014	0.055	< 0.0001	< 0.0001	18/20	no	3.79
	ViT-S/16	74.96 ± 12.20	82.32	+8.30 [−0.19, 16.80]	0.055	0.126	< 0.0001	< 0.0001	18/20	no	4.68
all test images	EfficientNet-B0	75.19 ± 11.20	75.87	–	–	–	–	–	–	–	5.15
	MobileNetV3-Large	73.68 ± 12.55	75.38	−1.52 [−8.23, 5.19]	0.641	0.641	0.2055	0.2305	8/20	no	5.49
	MobileNetV4-Conv-Medium	68.50 ± 10.90	70.47	−6.69 [−14.34, 0.96]	0.083	0.166	< 0.0001	< 0.0001	3/20	no	6.69
	ConvNeXt V2-Tiny	84.10 ± 9.83	84.88	+8.91 [1.65, 16.17]	0.019	0.075	< 0.0001	< 0.0001	19/20	no	3.51
	ViT-S/16	83.67 ± 8.69	84.59	+8.48 [0.18, 16.77]	0.046	0.137	< 0.0001	< 0.0001	19/20	no	4.94

ConvNeXt V2-Tiny minus ViT-S/16 (exploratory, video frames excluded): −0.15 pp [−5.65, 5.36], corrected $p = 0.956$.

Table S12. Per-class precision, recall and F1 (%, mean ± SD over runs in which the class was present), backbone experiment, video frames excluded

Class	Backbone	Precision	Recall	F1	Runs
Algal leaf spot	EfficientNet-B0	73.4 ± 25.6	62.7 ± 20.4	64.4 ± 20.0	40
	MobileNetV3-Large	69.5 ± 24.3	61.2 ± 23.3	63.3 ± 23.0	40
	MobileNetV4-Conv-Medium	64.5 ± 27.9	53.0 ± 21.9	54.6 ± 21.2	40
	ConvNeXt V2-Tiny	74.1 ± 25.5	85.4 ± 13.0	77.0 ± 20.8	40
	ViT-S/16	76.3 ± 24.6	80.4 ± 16.3	76.1 ± 20.4	40
Leaf blight complex	EfficientNet-B0	67.8 ± 14.6	80.3 ± 16.2	72.0 ± 12.2	40
	MobileNetV3-Large	67.1 ± 15.0	74.8 ± 17.0	68.8 ± 12.1	40
	MobileNetV4-Conv-Medium	58.5 ± 14.7	70.7 ± 22.0	61.2 ± 15.0	40
	ConvNeXt V2-Tiny	83.7 ± 9.5	79.4 ± 14.9	80.7 ± 10.8	40
	ViT-S/16	81.0 ± 14.6	78.3 ± 13.2	78.6 ± 11.2	40
Phomopsis leaf spot	EfficientNet-B0	45.0 ± 45.0	28.5 ± 26.9	31.9 ± 31.7	32
	MobileNetV3-Large	39.0 ± 41.0	28.2 ± 32.4	30.9 ± 33.8	32
	MobileNetV4-Conv-Medium	46.4 ± 44.9	25.4 ± 25.9	30.6 ± 30.2	32
	ConvNeXt V2-Tiny	47.2 ± 46.7	33.6 ± 33.6	38.6 ± 38.0	32
	ViT-S/16	44.3 ± 44.4	41.5 ± 39.6	41.4 ± 40.8	32

Class	Backbone	Precision	Recall	F1	Runs
Root and collar rot	EfficientNet-B0	91.8 ± 17.6	99.7 ± 1.0	94.4 ± 12.8	34
	MobileNetV3-Large	87.0 ± 22.4	98.8 ± 2.3	90.3 ± 18.2	34
	MobileNetV4-Conv-Medium	78.3 ± 25.1	95.2 ± 17.1	83.9 ± 22.4	34
	ConvNeXt V2-Tiny	99.8 ± 1.4	97.0 ± 10.2	98.0 ± 6.6	34
	ViT-S/16	99.4 ± 2.0	100.0 ± 0.0	99.7 ± 1.0	34

Table S13. Recall (%) for *Phomopsis* leaf spot, still photographs against video frames, pooled over the 40 runs of each backbone (all test images)

Backbone	Still photographs	Video frames
EfficientNet-B0	42.9	96.2
MobileNetV3-Large	46.3	97.6
MobileNetV4-Conv-Medium	43.8	94.8
ConvNeXt V2-Tiny	58.4	97.8
ViT-S/16	71.7	98.9

Table S14. Referral to the grower: the threshold on the maximum softmax probability is chosen on the inner-validation images only (smallest threshold at which the accepted validation images reach the target accuracy; all images are referred if none does)

Coverage: share of test images decided by the model. *Accuracy decided*: mean over runs with at least one decided image (pooled over images in brackets). *Accuracy, no referral*: accuracy if every image were decided (pooled). *Below target*: runs in which the accuracy on decided images was below the target. *All decided*: runs in which every test image was decided. Referral shares are means over runs in which the class was present.

Scoring	Target	Backbone	Coverage (%)	Accuracy decided (%)	Accuracy, no referral (%)	Below target	All decided	Referred: Algal	Referred: Leaf blight	Referred: *Phomopsis*	Referred: Root rot
video frames excluded (primary)	90 %	EfficientNet-B0	65.0 ± 33.5	83.3 ± 12.1 (79.0)	70.8	22/39	3/40	42.3	33.1	51.6	10.3
		MobileNetV3-Large	54.3 ± 34.4	84.4 ± 12.1 (78.6)	69.8	24/38	2/40	50.6	49.7	57.8	16.7
		MobileNetV4-Conv-Medium	57.8 ± 30.8	79.4 ± 13.3 (73.6)	64.8	29/37	0/40	48.8	41.7	62.7	16.7
		ConvNeXt V2-Tiny	89.1 ± 18.9	86.4 ± 9.0 (83.6)	80.5	21/40	9/40	12.0	11.5	14.4	3.3
		ViT-S/16	87.5 ± 17.8	86.1 ± 7.7 (84.6)	81.1	24/40	8/40	12.8	14.0	12.3	0.3
video frames excluded (primary)	95 %	EfficientNet-B0	52.5 ± 32.0	87.2 ± 11.0 (81.5)	70.8	29/39	2/40	53.9	48.7	65.4	13.2
		MobileNetV3-Large	39.8 ± 28.3	90.4 ± 11.0 (85.2)	69.8	19/37	0/40	68.8	65.8	78.8	20.6
		MobileNetV4-Conv-Medium	44.9 ± 27.4	84.0 ± 13.1 (77.1)	64.8	27/37	0/40	64.3	56.3	76.7	18.4
		ConvNeXt V2-Tiny	83.2 ± 23.9	88.4 ± 8.5 (85.6)	80.5	30/40	8/40	18.2	19.4	22.8	5.0
		ViT-S/16	75.9 ± 25.7	89.6 ± 7.3 (86.7)	81.1	29/40	4/40	24.7	27.2	29.4	0.5
all test images	90 %	EfficientNet-B0	72.6 ± 27.8	84.0 ± 10.9 (80.9)	73.7	22/39	3/40	37.0	26.4	22.2	3.3
		MobileNetV3-Large	61.1 ± 33.3	85.6 ± 11.2 (80.3)	73.0	23/39	2/40	44.3	43.1	37.7	16.4

Scoring	Target	Backbone	Coverage (%)	Accuracy decided (%)	Accuracy, no referral (%)	Below target	All decided	Referred: Algal	Referred: Leaf blight	Referred: *Phomopsis*	Referred: Root rot
		MobileNetV4-Conv-Medium	67.0 ± 28.0	81.1 ± 11.7 (76.2)	68.2	28/38	0/40	42.3	35.7	25.1	12.7
		ConvNeXt V2-Tiny	90.8 ± 16.4	88.0 ± 8.1 (85.2)	82.5	20/40	8/40	11.5	10.6	7.1	3.3
		ViT-S/16	92.4 ± 9.7	87.0 ± 7.2 (85.5)	83.1	22/40	10/40	9.3	9.7	3.8	0.3
all test images	95 %	EfficientNet-B0	60.8 ± 30.2	87.6 ± 10.5 (82.8)	73.7	25/39	2/40	49.0	41.6	34.0	10.9
		MobileNetV3-Large	47.5 ± 30.3	89.7 ± 10.7 (85.3)	73.0	22/39	1/40	62.8	60.1	51.8	17.9
		MobileNetV4-Conv-Medium	56.7 ± 26.9	84.7 ± 12.3 (78.6)	68.2	26/38	0/40	55.7	47.6	31.9	16.1
		ConvNeXt V2-Tiny	86.1 ± 21.6	89.5 ± 7.5 (86.8)	82.5	31/40	8/40	17.3	16.4	11.3	3.4
		ViT-S/16	82.3 ± 21.9	89.8 ± 6.8 (87.6)	83.1	33/40	5/40	20.3	22.6	11.0	0.5

Table S15. Threshold-free confidence ranking: accuracy on the most confident images at fixed coverage, area under the risk–coverage curve (AURC, lower is better) and top-2 accuracy (%, mean over 40 runs; video frames excluded)

Backbone	50 % coverage	70 % coverage	80 % coverage	100 % (accuracy)	AURC (mean ± SD)	Top-2 accuracy (mean ± SD)
EfficientNet-B0	87.3	80.9	78.5	72.8	13.27 ± 8.91	93.48 ± 5.09
MobileNetV3-Large	86.4	81.2	77.5	72.1	13.80 ± 9.20	92.83 ± 5.83
MobileNetV4-Conv-Medium	82.6	75.7	73.1	66.5	17.80 ± 9.77	91.60 ± 5.86
ConvNeXt V2-Tiny	94.3	90.7	88.8	82.9	6.47 ± 6.80	96.90 ± 2.59
ViT-S/16	94.2	91.2	89.2	82.3	6.32 ± 5.82	96.13 ± 2.33

Exploratory paired comparisons with EfficientNet-B0 over 20 folds (corrected resampled *t*-test, Holm over four comparisons; no test of these quantities was pre-specified):

Backbone	AURC difference [95% CI]	*p*	*p*, Holm	Top-2 difference (pp) [95% CI]	*p*	*p*, Holm
MobileNetV3-Large	+0.53 [−6.67, 7.73]	0.879	0.879	−0.65 [−3.65, 2.35]	0.655	0.815
MobileNetV4-Conv-Medium	+4.53 [−2.62, 11.68]	0.201	0.401	−1.88 [−6.12, 2.35]	0.364	0.815
ConvNeXt V2-Tiny	−6.80 [−12.54, −1.07]	0.023	0.079	+3.42 [−1.61, 8.45]	0.171	0.683
ViT-S/16	−6.95 [−12.67, −1.24]	0.020	0.079	+2.64 [−2.25, 7.54]	0.272	0.815

Table S16. Rerun agreement, model size and CPU latency

EfficientNet-B0 was retrained in the backbone experiment (same folds, seeds and schedule; `drop_last` in the training loader, as required by MobileNetV4). Its mean macro F1 was 66.65 % (video frames excluded) against 66.60 % in the first experiment; the mean absolute difference between matching runs was 3.02 points (correlation 0.93). On all test images the values were 75.19 % and 75.06 % (mean absolute difference 2.63 points).

Backbone	Parameters (M)	CPU latency, 1 thread, median (ms)	p90 (ms)	Mean training time per run (min)
EfficientNet-B0	4.013	36.71	50.15	2.74
MobileNetV3-Large	4.207	29.05	32.63	2.65
MobileNetV4-Conv-Medium	8.440	43.25	68.56	2.72
ConvNeXt V2-Tiny	27.870	163.44	219.13	2.96
ViT-S/16	21.667	100.82	113.68	2.63

Table S17. Sensitivity to the grouping of video frames: backbone comparison on test images whose video-linked group was not split

Mean test images per fold: 87.0. Corrected resampled *t*-test over 20 folds; Holm adjustment over the four comparisons.

Backbone	Macro F1 (%), mean ± SD	Difference from EfficientNet-B0 (pp) [95% CI]	*p*, corrected	*p*, Holm	Folds higher
EfficientNet-B0	67.00 ± 13.29	–	–	–	–
MobileNetV3-Large	63.25 ± 12.51	−3.74 [−13.10, 5.61]	0.413	0.413	6/20
MobileNetV4-Conv-Medium	56.87 ± 14.10	−10.13 [−21.07, 0.81]	0.068	0.149	3/20
ConvNeXt V2-Tiny	74.04 ± 13.94	+7.04 [0.68, 13.40]	0.032	0.127	18/20
ViT-S/16	73.82 ± 14.15	+6.82 [0.01, 13.64]	0.050	0.149	18/20

Table S18. Optimism of the inner-validation images used for the referral threshold (video frames excluded)

Mean accuracy without any referral on the validation images and on the test folds (means of per-run values), and the number of runs (of 40) in which the validation accuracy was already at or above the target.

Backbone	Validation accuracy (%)	Test accuracy (%)	Runs with validation accuracy ≥ 90 %	Runs with validation accuracy ≥ 95 %
EfficientNet-B0	81.0	72.8	16	11
MobileNetV3-Large	77.7	72.1	11	4
MobileNetV4-Conv-Medium	79.8	66.5	11	5
ConvNeXt V2-Tiny	93.2	82.9	32	25
ViT-S/16	91.9	82.3	29	18

Table S19. Absolute and referral results on the strict subset (test and validation images whose video-linked group lies entirely inside the evaluated subset)

Mean test images per run 87.0 (primary scoring: 121.8); mean validation images per run 34.2 (primary scoring: 34.8). This subset contains 1.4 % *Phomopsis* images against 19.3 % in the primary scoring, so it is cleaner but easier and is not a corrected estimate. Means over 40 runs; decided = images whose confidence reached the threshold chosen on the validation images.

Backbone	Macro F1 (%)	Accuracy (%)	AURC	Top-2 (%)	Accuracy at 70 % coverage (%)	Decided, 90 % target (%)	Accuracy on decided (%)	Runs below 90 %	Decided, 95 % target (%)	Accuracy on decided (%)	Runs below 95 %
EfficientNet-B0	67.0	77.0	10.57	95.5	84.9	69.3	84.9	20/39	57.1	88.8	28/39
MobileNetV3-Large	63.3	74.4	12.63	94.3	83.8	57.2	84.6	24/38	43.3	90.4	19/37
MobileNetV4-Conv-Medium	56.9	68.7	16.83	93.9	78.3	61.4	79.9	29/38	48.8	84.0	28/38
ConvNeXt V2-Tiny	74.0	86.8	4.21	97.4	94.1	89.8	90.0	14/40	84.8	91.1	28/40
ViT-S/16	73.8	85.2	4.54	96.7	93.8	88.3	89.2	18/40	76.9	92.3	24/40

Table S20. Top-2 hit rate and errors among automatically decided images, by true class (primary scoring, video frames excluded)

Pooled over the 40 runs of each backbone (images, not run means). Top-2: the correct class was among the two most probable. Decided: confidence reached the threshold set on the validation images for a 90 % target. Error among decided: share of decided images of that class that were misclassified. *Phomopsis* images come essentially from one capture group.

Backbone	Class	Test images	Recall (%)	Top-2 (%)	Decided (%)	Error among decided (%)
EfficientNet-B0	Algal leaf spot	1620	63.9	92.8	61.1	29.4
EfficientNet-B0	Leaf blight	1530	80.7	98.7	63.0	16.9
EfficientNet-B0	Phomopsis leaf spot	940	42.9	74.0	35.4	54.1
EfficientNet-B0	Root and collar rot	780	99.2	99.7	94.4	0.1
MobileNetV3-Large	Algal leaf spot	1620	64.6	92.4	54.6	28.2
MobileNetV3-Large	Leaf blight	1530	75.9	96.9	52.9	21.1
MobileNetV3-Large	Phomopsis leaf spot	940	46.3	72.7	38.9	42.9
MobileNetV3-Large	Root and collar rot	780	97.2	99.5	86.7	1.2
MobileNetV4-Conv-Medium	Algal leaf spot	1620	57.5	90.2	55.8	34.2
MobileNetV4-Conv-Medium	Leaf blight	1530	69.5	98.1	58.3	25.9
MobileNetV4-Conv-Medium	Phomopsis leaf spot	940	43.8	69.4	42.7	52.6
MobileNetV4-Conv-Medium	Root and collar rot	780	95.9	97.8	90.1	2.3
ConvNeXt V2-Tiny	Algal leaf spot	1620	87.2	98.3	91.4	9.9
ConvNeXt V2-Tiny	Leaf blight	1530	77.6	97.8	88.4	20.3
ConvNeXt V2-Tiny	Phomopsis leaf spot	940	58.4	89.0	81.4	38.3
ConvNeXt V2-Tiny	Root and collar rot	780	99.1	99.6	99.1	0.5
ViT-S/16	Algal leaf spot	1620	81.9	97.2	89.1	14.3
ViT-S/16	Leaf blight	1530	76.4	96.3	85.4	20.5
ViT-S/16	Phomopsis leaf spot	940	71.7	89.3	80.9	24.3
ViT-S/16	Root and collar rot	780	100.0	100.0	99.9	0.0